Reg. No:.....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION ....

MODEL QUESTION PAPER - THEORY

Fourth Semester

MG4DSCSTA202- Probability Distributions and Statistical Inference

(2024 ADMISSION)

Duration: 1 Hour 30 minutes *Maximum Marks: 50*

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A (Short Answer Questions)

*Answer any **seven** questions. Each question carries **2 marks**.*

1. Define hyper geometric distribution. [K][1]

2. How negative binomial distribution is related with geometric distribution? [K][1]

3. State additive property of gamma distribution. [U][1]

4. Define type II beta distribution and state its HM. [U][1]

5. Give a situation in which lognormal distribution can be applied. How it is connected with normal distribution? [K][1]

6. Give any two advantages of Tchebychev's inequality. [K][2]

7. Let $\bar{X}$ be the mean of a sample of size 9 from a normal population with mean 5. Compute P($\bar{X} > 5$). [A][2]

8. What are the two types of errors in a statistical test procedure? [K][2]

9. What are the assumptions used in F test? [U][3]

10. A die is thrown 9000 times and a throw of 3 or 4 was observed 3240 times. Check whether the die was unbiased. [A][3]

(7x2=14)

Part B (Short Essay Questions)

*Answer any **four** questions. Each question carries **6 marks.***

11. Rahul likes to play cards. He drew six cards from a pack of 52 cards without replacement. What is the probability that there will be three black cards in the selected six cards? [U][1]

12. Suppose a telemarketer has a 15% chance of making a sale on any given phone call. He is required to make 3 successful sales before 2.00pm in a particular day. (a) What is the probability that he needs to make 7 calls to ensure the third sale? (b) Compute the expected number of phone calls to ensure five sales. [U][1]

13. Discuss the suitable transformation that converts a type I beta variate to a type II beta variate. [U][1]

14. It is known that on the average 2/3 of the seeds of a certain plant germinate. Use Tchebychev's inequality to obtain an upper bound to the probability that the number germinating will differ from the expected number by more than 10 if 100 seeds are planted. [A][2]

15. Explain t- tests used in different situations of statistical analysis. [U][3]

16. Explain (a) sign test and (b) Wilcoxon signed rank test. [U][3]

(4x6=24)

Part C (Essay Type Questions)

*Answer any **one** question. Each question carries 12 **marks.***

17. (a) State and prove weak law of large numbers. (b) Examine whether weak law of large numbers holds for the sequence $\{X_k\}$ of independent random variables defines as follows: P($X_k = k$)= P($X_k = -k$) = $\frac{1}{2}$. [A][2]

18. Explain Mann- Whitney U test. State its advantages and limitations. [U][3]

(1x12=12)